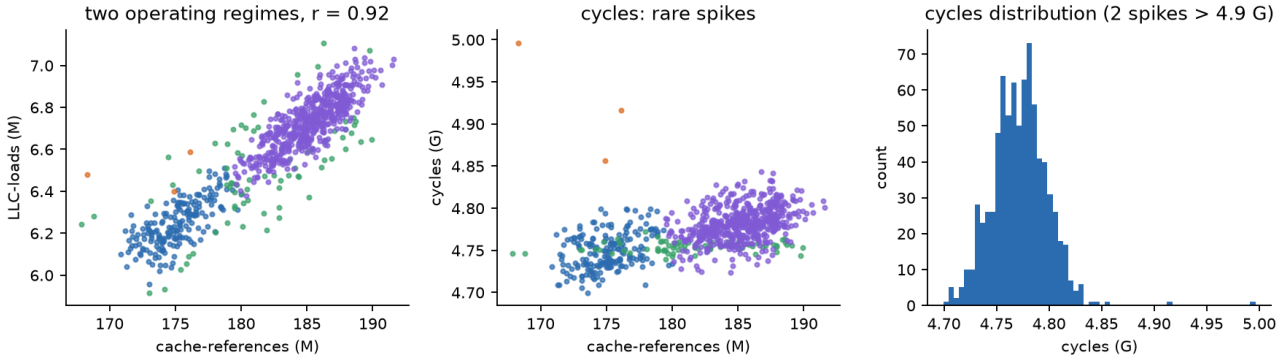
Detecting neural-network backdoor inferences from hardware performance counters

Strict one-class anomaly detection — ensemble of density models with out-of-fold calibration

1. Problem and what the data allows

We receive 800 HPC traces (cache-references, cycles, LLC-loads) from **clean** inferences only and must label unseen traces as clean or backdoor. No backdoor example exists, so any supervised classifier is impossible; the problem is novelty detection. The private set will be scored with Accuracy, TPR, FPR, F1 and AUROC, so we need *both* a good ranking (AUROC) and a well-calibrated decision threshold.

Findings from the clean file (Fig. 1): (i) the counters are strictly positive and multiplicative, so we model **log-counters** (a ratio such as LLC/cache-ref becomes a difference); (ii) cache-references and LLC-loads are highly correlated (r = 0.92) and the data form **multiple operating regimes** (BIC picks 4 Gaussian components; a single Gaussian is the wrong shape); (iii) rows are independent (lag-1 autocorrelation ≈ 0), so no temporal model is needed; (iv) cycles has rare OS-noise spikes (two rows at 6–9 robust sigma), which must not be learned as “normal” nor cause a flood of false positives.



2. Method

Pipeline. raw counts → log → winsorised whitening (ZCA) → three one-class detectors fitted on clean data only → tail calibration → evidence fusion → threshold.

Detector	Models	Why it is in the ensemble
GMM (BIC-selected k)	negative log-likelihood of the clean density	matches the multi-regime shape; best single model
kNN (k = 10)	mean distance to nearest clean traces	non-parametric; no distributional assumption
Isolation Forest	random-partition isolation depth on whitened features	independent inductive bias; axis-aligned view

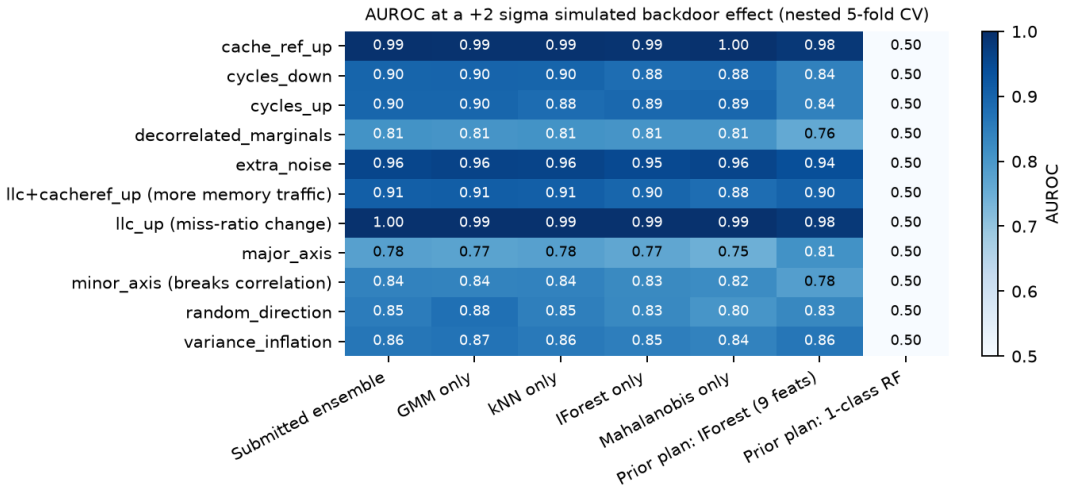
Calibration (the part that fixes accuracy / F1 / FPR). Scores from different detectors are not comparable, and in-sample scores of the training rows are over-optimistic. We therefore compute *repeated 5-fold out-of-fold* clean scores (3 repeats) and, per detector, map any score s to $-\log P_{clean}(S \geq s)$: empirical survival function in the bulk and an exponential (GPD, ξ=0) tail above the 90th percentile, so very strong anomalies keep ranking above moderate ones instead of saturating at 1/n. The three evidence values are averaged (a Fisher-style fusion that rewards any detector that sees something). The decision threshold is the 95th percentile of the fused *out-of-fold* clean evidence, so the clean-FPR is controlled *by construction* at 5% and does not depend on a guessed contamination rate. Final models are then refit on all 800 traces.

Robustness details. columns are selected by name (order and extra columns such as a label are ignored); rows with NaN/zero/negative counters cannot be modelled and are returned as anomalous instead of crashing; whitening is winsorised (1%/99%) so the two cycle spikes do not distort the metric; everything is seeded and covered by unit tests.

3. Validation protocol — and its honest limit

Because there are no backdoor traces, **we cannot measure real TPR**. Rather than hand-picking a single synthetic anomaly (e.g. “LLC×1.3”), we benchmark against **11 families of plausible trigger effects** at 1, 2 and 3 clean standard deviations: single-counter shifts up/down, miss-ratio change (LLC only), more memory traffic (LLC and cache-ref together), shifts along the major and minor principal axes (the minor axis breaks the correlation structure while keeping

marginals normal), random directions, variance inflation, independent column shuffling (marginals intact, dependence destroyed) and extra noise. A **nested** 5-fold protocol is used: the entire pipeline, including its inner out-of-fold calibration and threshold, is fitted on 4/5 of the clean data and tested on the held-out clean fifth (negatives) versus a perturbed copy (positives). These numbers are a relative yardstick for choosing methods and thresholds, not a forecast of the private-set score.



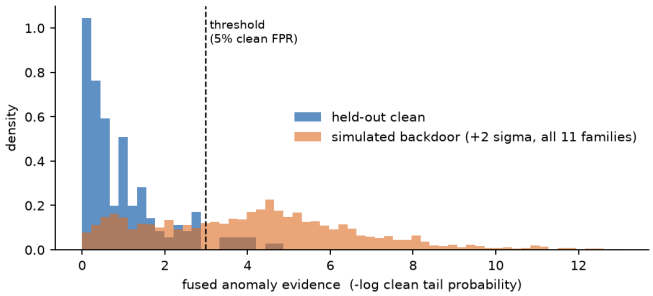
Method	mean AUROC	worst-family AUROC	TPR @ 5% FPR
Submitted ensemble (GMM+kNN+IForest)	0.867	0.788	0.631
GMM only	0.869	0.797	0.640
kNN only	0.866	0.790	0.629
LOF only	0.859	0.706	0.616
Isolation Forest only	0.853	0.759	0.557
Robust Mahalanobis only	0.851	0.739	0.563
One-class SVM only	0.831	0.735	0.634
Prior plan: IForest on 9 raw/ratio/log features	0.846	0.761	0.537
Prior plan: Random Forest trained on one class	0.500	0.500	0.000

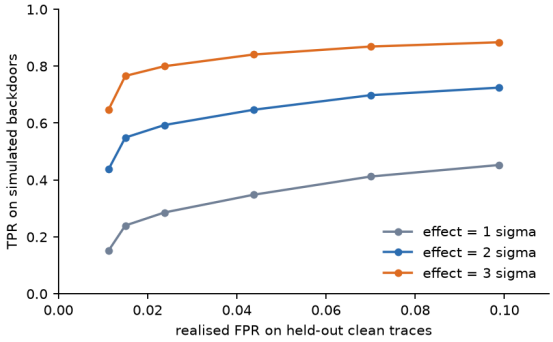
Mean over 11 families × 3 magnitudes × 5 folds. The best single model (GMM) and the ensemble are statistically tied, on both mean and worst-family AUROC. The ensemble is kept as insurance: the per-family winner among GMM and kNN changes (GMM wins 6 families, kNN 5), and a single model can fall behind on an effect we did not simulate (e.g. Mahalanobis reaches only 0.84 on “more memory traffic” where kNN reaches 0.88). The single-class Random Forest in the earlier plan outputs a constant score (AUROC exactly 0.5): a classifier cannot flag a class it has never seen.

4. Operating point

The fused-evidence threshold is set so that the realised clean FPR equals the target. The check below is on held-out clean folds with the full nested pipeline: realised FPR tracks the target (slightly conservative), and accuracy / F1 flatten out beyond about 5–8% FPR, so we deploy **5%**. If the private set contains a much smaller share of backdoor traces, lower it (`predict.py --target-fpr 0.02`); no retraining is needed.

target FPR	realised FPR	TPR (all effects)	TPR (>= 2 sigma)	Acc (50/50, >= 2 sigma)	F1 (>= 2 sigma)
1%	0.011	0.412	0.542	0.766	0.651
2%	0.015	0.518	0.657	0.821	0.757
3%	0.024	0.559	0.696	0.836	0.786
5%	0.044	0.612	0.744	0.850	0.815
8%	0.070	0.660	0.783	0.857	0.832
10%	0.099	0.687	0.804	0.853	0.834





5. Limitations and how to use

• Real backdoor traces may differ from our simulated families; effects below ~1 sigma in all counters are fundamentally undetectable from three tightly clustered counters (AUROC ≈ 0.75 at 1 sigma). Strong effects (≥3 sigma) are detected with AUROC ≈ 0.95.

• If the private file contains extra HPC columns, the detector uses only the three it was trained on; to use more counters, retrain with `HPCBackdoorDetector(columns=[...])`.

• Commands: `python train.py` (fits and saves models/detector.joblib), `python predict.py --input file.csv` (scores; prints Accuracy/TPR/FPR/F1/AUROC if a label column is present), `python evaluate.py` (reproduces every number here), `python make_report.py`.